

Alan M. Tommy

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SUMMARY

Computer science graduate with experience spanning full-stack development, data analytics, and artificial intelligence. Self-motivated and versatile, proven to deliver under pressure across diverse technical domains and team environments.

EDUCATION

University of Michigan, Ann Arbor, MI

Bachelor of Science in Engineering, Computer Science – May 2025

GPA: 3.74/4.00

Key Courses

Statistics and Artificial Intelligence, Software Engineering, Operating Systems, Web Systems, Computer Security, Machine Learning, Data Analytics Tools and Techniques, Computer Game Design and Development

EXPERIENCE

University of Michigan ROTC – Wargame Digitalization Team Intern

June 2025 – January 2026

- Worked in a team to develop a first-of-its-kind digital interface for a strategic military simulation on the web.
- Wrote more than 25% of the entire codebase, implementing PostgreSQL database, Django backend, Redis-backed WebSockets, user authentication/authorization, and several frontend React components.
- Application has been demonstrated to successfully support **over 100 concurrent users and is being used to train and evaluate future officers** from the Army, Navy, Marine Corps, Air Force, and Space Force.

PROJECTS

Final Fumes (Video Game)

February 2025 – April 2025

- Developed a fully-fledged video game in Unity on a 4-person team within an 8-week Agile development cycle.
- Devised a procedural generation algorithm inspired by graph-based algorithms to spawn terrain while ensuring logical road connectivity.
- Showcased to live players; received overwhelmingly positive feedback, placing **4th among 20+ teams**.

Multi-Threaded Network File Server

November 2024 – December 2024

- Implemented a multi-threaded, crash-consistent network file server in C++20/Boost.Threads, handling concurrent client requests over raw TCP sockets.
- Enforced disk write ordering (data/inode blocks before referencing metadata) to guarantee on-disk file system consistency across crashes.
- Used fine-grained per-block locking to allow concurrent reads/writes to different disk blocks while serializing conflicting writes to the same block, and safely parsed untrusted client input over the network protocol.

Kernel Memory Manager

October 2024 – November 2024

- Designed and implemented a virtual memory pager in C++ that manages process address spaces, handling page faults, swap-backed and file-backed page mappings, and simulated MMU page table entries.
- Implemented the clock page replacement algorithm (FIFO with second-chance) with dirty/referenced bit tracking to minimize disk I/O on eviction.
- Built copy-on-write page sharing for fork()-based address space duplication, deferring physical page copies until write faults occur to reduce memory and I/O overhead.

SKILLS

Programming Languages: C++, C, Python, SQL, JavaScript, TypeScript, Visual Basic, C#, R, MATLAB, Lua

Tools and Software: Linux/Unix, GitHub, Docker, Blender, Vim, Unity, Figma, Jira

Machine Learning: Data preprocessing, model tuning, feature engineering, TensorFlow, PyTorch